Introduction to Stata

Lecture 1: Workflow and getting data into Stata

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Four lectures, one project

The plan

1. **Workflow and data in** Ch. 1–3
2. **Variables and combining data** Ch. 4–5
3. **Statistics, tests, graphs** Ch. 6–7
4. **Linear regression (OLS)** Ch. 8

Chapters refer to the handbook *Introduction to Stata*.

Handbook: Chapters 1–3.

The project

- Bank of Italy *Survey on Household Income and Wealth* (SHIW), 2010–2022
- How are household income and wealth related to education, age and geography?
- Each lecture starts from the files saved by the previous one

How each lecture works

45 minutes: demonstration

- Slides + commands run live in Stata
- Part 1 of `L1_exercises.do`: the same commands, run them with me

Break

60 minutes: exercises

- 5 exercises in Part 2 of `L1_exercises.do`: 10 minutes each, then we write the solution together in Stata
- Write your code below each task, in the same do-file

Files

- `L#_exercises.do`: commands of the demo + exercise tasks
- `L#_solutions.do`: everything solved and commented line by line, published after the lecture

Lost an exercise? The next lecture starts from the files saved by the solutions do-file: run it and you are back on track.

PART 1

Demonstration

Before we start: the project folder

Handbook § 1.2

```
Stata_Lectures_2026/
├── do/                  do-files
├── data/raw/            SHIW files go here
├── data/clean/          created by Stata
├── output/figures/
├── output/tables/
└── log/
```

Download (Bank of Italy, *Distribuzione dei microdati*)

- `storico_ascii.zip`: copy `comp, rfam, ricf, cons, peso, defl, rper` (`.csv`) into `data/raw/`
- The codebook (Excel): `documentazione_storico_versione_12.2.xlsx`, also into `data/raw/`

Rules

- Never modify the raw files
- Cite: *Banca d'Italia, Indagine sui bilanci delle famiglie italiane*, archive version 12.2

Getting the data (1): by hand

Handbook § 3.2

The microdata: `https://www.bancaditalia.it/statistiche/tematiche/indagini-famiglie-imprese/bilanci-famiglie/distribuzione-microdati/`

or: `www.bancaditalia.it` → *Statistiche* → *Indagini sulle famiglie e le imprese* → *Bilanci delle famiglie italiane* → *Distribuzione dei microdati*

1. Under *Archivio storico*, click **ASCII** (`storico_ascii.zip`, 42 MB)
2. Unzip it: you get a folder `CSV/` with 19 files
3. Copy `comp, rfam, ricf, cons, peso, defl, rper` (`.csv`) into `data/raw/`

The codebook: same section → *Documentazione*

4. Download `documentazione_storico_versione_12.2.xlsx` into `data/raw/`

Conditions of use (on the download page)

Distributed *for research purposes only*; use under the sole responsibility of the user; cite *Banca d'Italia, Indagine sui bilanci delle famiglie italiane* and the archive version; send references of works to `indagini.famiglie@bancaditalia.it`. Do not redistribute the files.

Getting the data (2): from Stata

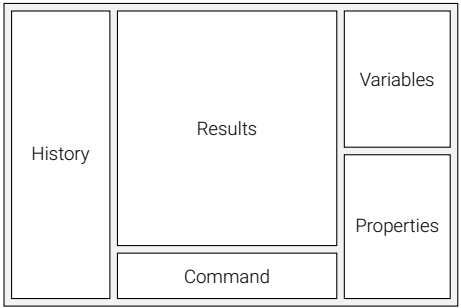
Handbook § 3.2

```
local bdi "https://www.bancaditalia.it/statistiche/tematiche/indagini-famiglie-imprese/bilanci-famiglie"
copy "‘bdi’/distribuzione-microdati/documenti/storico/storico_ascii.zip" ///
     "$raw/storico_ascii.zip", replace                // copy accepts a web address
cd "$raw"
unzipfile "storico_ascii.zip", replace                 // creates data/raw/CSV/
cd "$root"
foreach f in comp rfam ricf cons peso defl rper {
    copy "$raw/CSV/‘f’.csv" "$raw/‘f’.csv", replace
}
```

- The same steps as by hand, but reproducible: anyone can rerun them
- `L1_exercises.do` runs this block only if `comp.csv` is not already in `data/raw/`
- If the download fails (network, proxy): the do-file stops with a message; download by hand (previous slide)

The Stata interface

Handbook § 1.1



- **Command window**: try a command once
- **Do-file editor** (`doedit`): where the real work happens
- Run selected lines: Ctrl+D (Windows) / Cmd+Shift+D (Mac)
- **Data browser** (`browse`): look, never edit by hand

If it is not in a do-file, it did not happen.

A reproducible setup

Handbook § 1.2

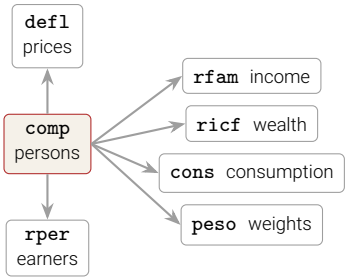
```
clear all                              // empty memory
set more off                           // no pauses in the output
version 19                             // syntax of Stata 19, forever

global root "C:/path/to/Stata_Lectures_2026"  // the ONLY line to change
global raw   "$root/data/raw"          // all other paths derive from root
global clean "$root/data/clean"
capture mkdir "$clean"                 // capture: no error if it already exists
cd "$root"

capture log close
log using "$root/log/L1.log", text replace   // a record of all output
```

Forward slashes `/` also on Windows. Avoid folders synchronised by OneDrive/iCloud if Stata cannot save.

The SHIW data: files and keys



File	Identified by
comp, rper	`nquest nord anno`
rfam, ricf, cons, peso	`nquest anno`
defl	`anno`

`nquest`: household
`nord`: person in the household
`anno`: survey wave

Households interviewed in several waves keep their `nquest`: a panel.

Importing and looking at the data

Handbook § 3.2, 3.3

```
import delimited "$raw/comp.csv", clear case(lower)   // CSV -> memory
describe                                               // variables, types, labels
list nquest nord anno sesso eta studio in 1/5          // first rows
summarize anno eta                                     // N, mean, sd, min, max
tabulate anno                                          // frequencies
```

- `import excel ..., sheet("...") firstrow` for Excel files (Exercise 5)
- `use` / `save` for Stata `.dta` files
- Stata is case-sensitive: `anno` and `ANNO` are different variables

The syntax of (almost) every Stata command

Handbook § 2.1, 2.6

```
[by varlist:]  command [varlist] [=exp] [if exp] [in range] [weight] [, options]
```

```
summarize eta                                  // one variable
summarize eta if anno == 2022                  // only some observations
summarize eta if anno == 2022, detail          // an option
list nquest anno eta in 1/5                    // first five rows
bysort anno: summarize eta                     // repeat for each wave
```

Commands can be abbreviated (`su`, `tab`, `gen`): the minimum is underlined in `help`.

Operators and the missing value trap

Handbook § 2.2, 2.3

`==`	equal (double!)
`!=`	not equal
`> >= < <=`	comparisons
`&`	and
`|`	or
`!`	not

A missing value (`.`) is larger than any number:




```
count if studio >= 4                    // 197,838: WRONG
count if studio >= 4 & !missing(studio) // 133,459
```

Keep, drop and identifiers

Handbook § 3.4, 3.8, 3.9

anno	nquest	nord	eta	sesso
2020	2886	1	71	1
2020	2886	2	68	2
2022	173	1	74	1
2022	2886	1	73	1
2022	2886	2	70	2

Rows from `comp.csv`: household 2886 is interviewed in 2020 and 2022.

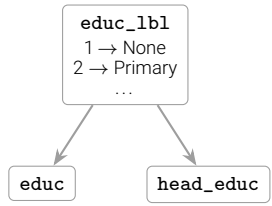
```
keep if anno >= 2010          // rows
keep nquest nord anno ...     // columns
isid nquest nord anno         // unique?
duplicates report nquest anno // not unique
```

- Always know what one row is
- `isid` stops the do-file if the key is not unique
- IDs as `long`, never `float` (7 digits)

Names and labels: the data contain only numbers

Handbook § 3.5

```
rename studio educ
label variable educ "Educational attainment"
label define educ_lbl 1 "None" 2 "Primary" ///
   3 "Lower secondary" 4 "Upper secondary"  ///
   5 "University degree" 6 "Postgraduate"
label values educ educ_lbl
tabulate educ            // shows the text
tabulate educ, nolabel   // shows the codes
```



A value label is a dictionary: define it once, attach it to many variables. The codes come from the codebook.

Check and save

Handbook § 3.1, 3.4, 3.6

```
compress                                    // smallest storage types
assert inlist(sex, 1, 2)                    // stop if false somewhere
assert age >= 0 & age < 120
save "$clean/individuals.dta", replace
```

You will also need (exercises):

`strtrim()`, `strupper()`, `substr()`	clean text (§ 2.4)
`usubinstr(s, old, new, .)`	replace text (§ 2.4)
`encode`, `destring`, `tostring`	strings ↔ numbers (§ 3.6)

Warning: `encode` numbers categories in alphabetical order: never use it as a key to match files.

Commands of the day

`clear all`, `version`	clean start	`keep`, `drop`	rows or columns
`global`, `cd`	paths	`isid`, `duplicates`	identifiers
`log using`	record output	`sort`, `order`	arrange
`help`	documentation	`rename`	names
`import delimited`	CSV files	`label var/define/values`	labels
`import excel`	Excel files	`compress`, `assert`	check
`describe`	structure	`save`, `use`	`.dta` files
`list`, `browse`, `count`	look	`encode`, `strtrim()`	text
`summarize`, `tabulate`	first statistics	`display`	calculator

PART 2

Now let's practice on Stata

Open `L1_exercises.do`, Part 2: five exercises, 10 minutes each,
then we write the solution together.

Before Lecture 2

Run `L1_solutions.do` (published after this lecture). It rebuilds from the raw data the files that Lecture 2 starts from. If you missed earlier lectures, run all the solutions do-files from Lecture 1, in order.